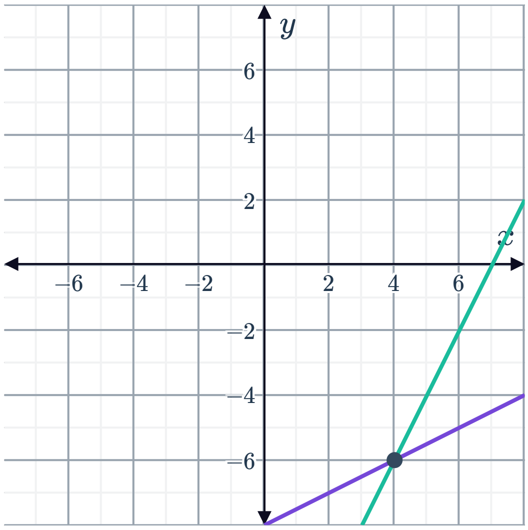
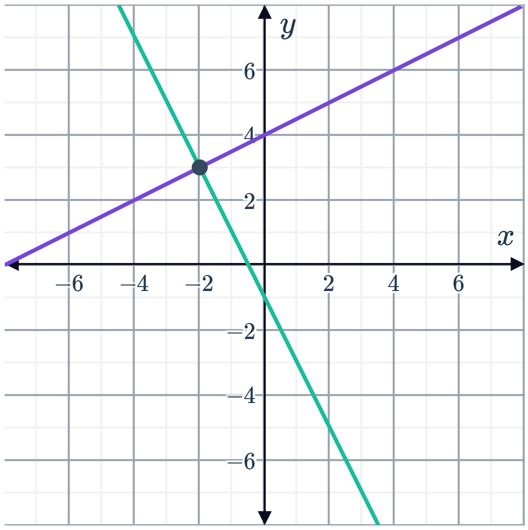
Graphical solutions

1 Each graph below shows a system of two equations. Write down the solution to each system, as a point in the form (x, y).

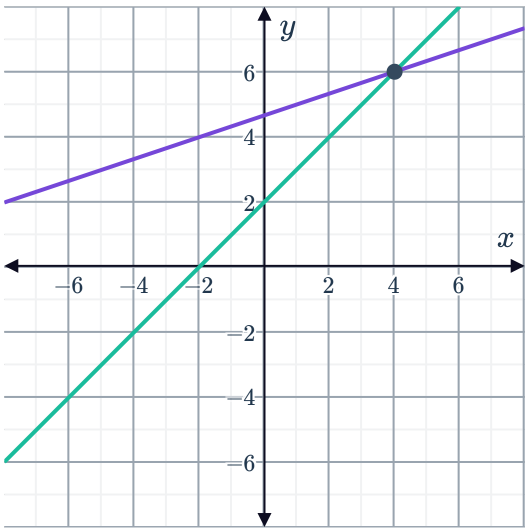
a



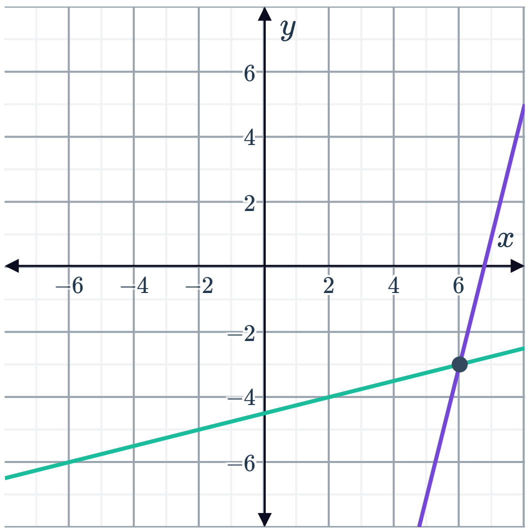
b



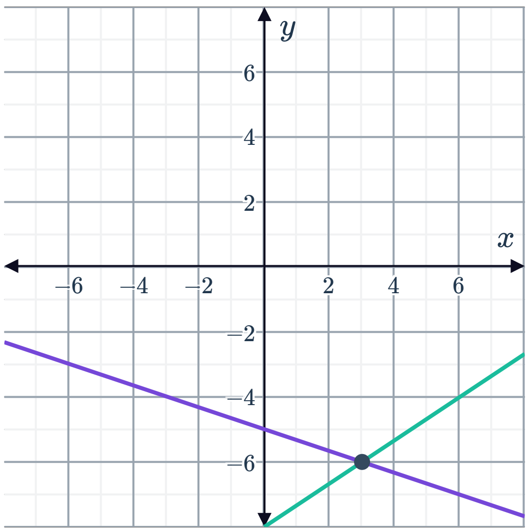
c



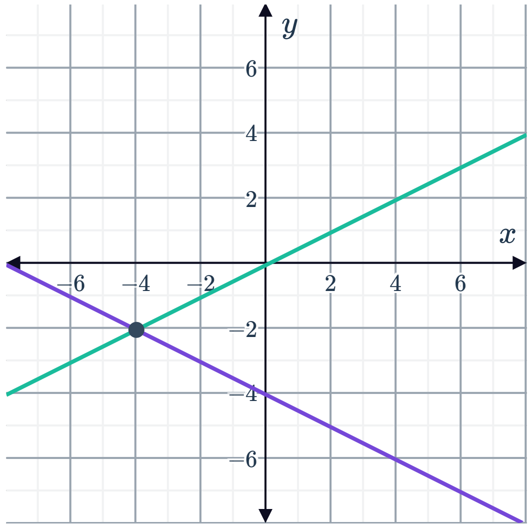
d



e



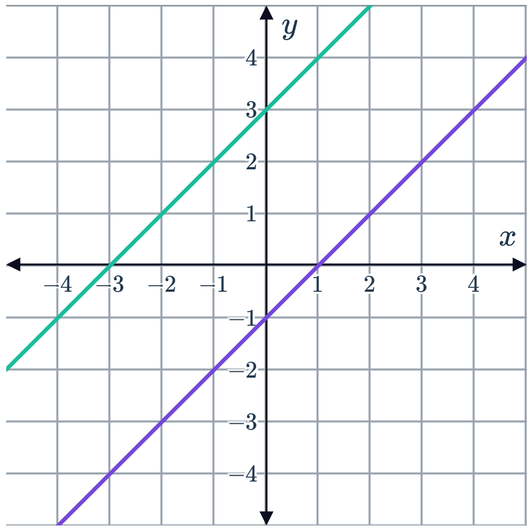
2 Consider the graph showing a system of two equations:



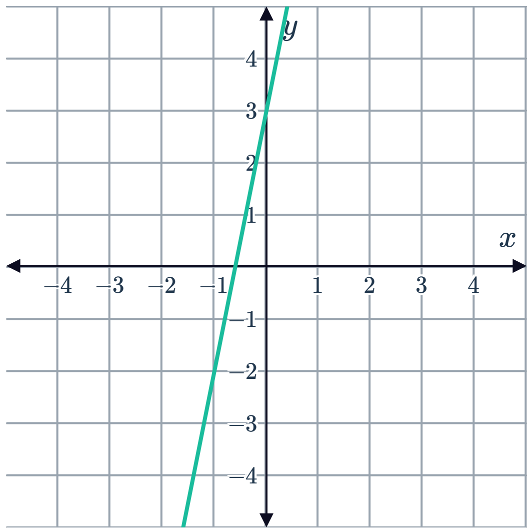
a How many solutions does this system of equations have?

b Write down the solution to the system of equations as an ordered pair (x, y).

3 Consider the graph showing a system of two equations:
How many solutions does this system of equations have?



4 Consider the graph of the equation $y = 5x + 3$:
If a second line $y = mx + b$ intersects this line at the point $(0, 3)$, state the value of b.



5 Consider the following systems of linear equations:

i Find the x and y-intercepts of the first equation.

ii Find the x and y-intercepts of the second equation.

iii Graph the two equations on the same set of axes.

iv State the coordinate pair (x, y) that satisfies both equations.

a $y = x - 1$
$y = -2x + 8$

b $y = x + 1$
$y = 4x + 4$

c $y = x + 3$
$y = 3x + 3$

d $y = -2x + 2$
$y = x - 7$

e $y = -x - 1$
$y = -3x - 3$

f $y = 2x + 2$
$y = -x + 2$

6 Consider the following linear equations:

$$y = 2x + 2$$
$$y = -2x + 2$$

a Determine the gradient and y-intercept of the line $y = 2x + 2$.

b Determine the x and y-intercepts of the line $y = -2x + 2$.

c Hence sketch the graphs on the same coordinate plane.

d State the values of x and y which satisfy both equations.

7 Consider the following system of equations: $y = 4x - 3$
$y = 4 - 3x$

a Complete the table of values for $y = 4x - 3$:

x	-3	-2	-1	0	1	2	3
y							

b Complete the table of values for $y = 4 - 3x$:

x	-3	-2	-1	0	1	2	3
y							

c Find the gradient of $y = 4x - 3$.

d Find the gradient of $y = 4 - 3x$.

e Graph the two equations on the same set of axes.

f Using your answer from part (e), find the point of intersection of the lines.

8 Consider the following system of equations:
$$y = x - 2$$
$$y = -3x - 6$$

a Find the gradient, and the y-value of the y-intercept, of the line $y = x - 2$.

i gradient

ii y-value of the y-intercept

b Find the gradient, and the y-value of the y-intercept, of the line $y = -3x - 6$.

i gradient

ii y-value of the y-intercept

c Graph the two equations on the same set of axes.

d State the coordinate pair (x, y) that satisfies both equations.

9 Consider the following system of equations:
$$y = -4x - 1$$
$$y = -4x + 2$$

a Graph the two equations on the same set of axes.

b Does there exist a value for x and y that satisfy the following two equations simultaneously?

10 For each of the following systems of equations:

i Graph the two equations on the same set of axes.

ii State the values of x and y that satisfy both equations.

a
$$y = x$$
$$y = -1$$

b
$$4x - 2y = 2$$
$$-2x + 4y = 2$$